Technical Diagnosis of the L2D-SLDS Paper and Implementation

Repository audit written from `Paper/icml2026/main.tex` and local code

2026-03-25

Abstract

This document is a rigorous diagnosis of the research paper in `Paper/icml2026/main.tex` and the accompanying implementation in the repository root. The goal is not to summarize at a high level, but to reconstruct the exact mathematical object being proposed, identify what is and is not formally defined, map the implementation back to the paper line by line, and isolate paper errors, code errors, hidden assumptions, and protocol mismatches. Every important criticism below is justified by direct references to the paper or code.

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1 Paper Overview

1.1 Problem setting

The paper studies sequential expert routing under partial feedback, non-stationarity, and dynamic expert availability. The formal setup is given in `Paper/icml2026/Section/Approach.tex:10-106`. At round t:

- the environment reveals a context $x_t \in \mathcal{X} \subseteq \mathbb{R}^d$,
- a target $y_t \in \mathcal{Y} \subseteq \mathbb{R}^{d_y}$ exists,
- a non-empty available expert set $\mathcal{E}_t$ is observed,
- the router selects $I_t \in \mathcal{E}_t$,
- only the queried prediction $\hat{y}_{t,I_t}$ and target y_t are observed,
- therefore only the queried residual

$$e_t = e_{t,I_t} = \hat{y}_{t,I_t} - y_t$$

  and queried cost C_{t,I_t} are observed.

  The paper defines the potential residual and cost of expert k at time t as

$$e_{t,k} = \hat{y}_{t,k} - y_t, \qquad C_{t,k} = \psi(e_{t,k}) + \beta_k,$$

where ψ is a convex loss and $\beta_k \geq 0$ is a known consultation fee; see `Paper/icml2026/Section/Approach.tex:24-4`

The paper allows a time-varying expert registry $\mathcal{K}_t$ that may be larger than the currently available set $\mathcal{E}_t$, and stale experts may be pruned from the registry; see `Paper/icml2026/Section/Approach.tex:17-20` and `Paper/icml2026/Section/Approach.tex:343-447`.

1.2 Main claim

The main claim is that a factorized switching linear dynamical system over expert residuals yields a tractable predictive belief over *unqueried* experts under censoring, and that this can be used for routing by minimizing a myopic IDS-style information ratio. This claim is developed in `Paper/icml2026/Section/Approach.tex:108-340`. The latent structure has:

- a regime variable z_t,
- a shared factor g_t,
- expert-specific states $u_{t,k}$,
- context-dependent regime transitions $\Pi_\theta(x_t)$,
- a residual emission model

$$e_{t,k} \mid z_t, g_t, u_{t,k}, x_t \sim \mathcal{N}\Big(\Phi(x_t)^\top (B_k g_t + u_{t,k}), R_{z_t,k}\Big).$$

The claimed benefit of the shared factor is cross-expert information transfer: after querying one expert, posterior beliefs about other experts change because their residual laws share dependence on g_t.

1.3 Claimed contributions

The claimed contributions are listed in `Paper/icml2026/Section/Introduction.tex:28-36`:

1. a formulation of online expert routing for non-stationary time series with bandit feedback and dynamic availability;
2. the L2D-SLDS model;
3. an IMM-style filtering recursion and pruning mechanism;
4. an IDS-inspired routing rule based on information about (z_t, g_t);
5. empirical evaluation on synthetic, Melbourne, and FRED tasks.

Assessment of the contribution statement. This list is internally coherent. However, it mixes three different levels of claim:

- formal problem definition,
- algorithm design,
- empirical evidence.

What it does *not* provide is a learning-theoretic guarantee such as regret or consistency. The framing invokes online learning and bandits, but the paper's formal guarantee layer is much weaker than the framing suggests.

2 Precise Reconstruction of the Method

2.1 Formal setup

I separate here what the paper states explicitly from what is only implied.

Explicitly stated. The decision-time sigma-algebra is

$$\mathcal{F}_t = \sigma(\mathcal{H}_{t-1}, x_t, \mathcal{E}_t),$$

with $\mathcal{H}_t = ((x_\tau, \mathcal{E}_\tau, O_\tau))_{\tau=1}^{t}$ and $O_t = (I_t, \hat{y}_{t,I_t}, y_t)$; see `Paper/icml2026/Section/Approach.tex:52-63`.

Explicitly stated.A policy $\pi_t(\cdot \mid \mathcal{F}_t)$ is an $\mathcal{F}_t$-measurable distribution over $\mathcal{E}_t$, and the action is sampled as $I_t \sim \pi_t(\cdot \mid \mathcal{F}_t)$; see `Paper/icml2026/Section/Approach.tex:59-63`.

Explicitly stated. The objective is expected cumulative cost

$$J(\pi) = \mathbb{E}\left[\sum_{t=1}^{T} C_{t,I_t}\right];$$

see `Paper/icml2026/Section/Approach.tex:90-106`.

Explicitly stated. As a one-step ideal benchmark, the paper defines

$$k_t^\star \in \arg\min_{k \in \mathcal{E}_t} \mathbb{E}[C_{t,k} \mid \mathcal{F}_t],$$

which it calls the myopic Bayes selector; see `Paper/icml2026/Section/Approach.tex:96-105`.

Missing. The paper does *not* define:

- a regret notion,
- a comparator class over policies,
- a non-stationary regret benchmark,
- a bandit sample complexity target,

- or a consistency criterion for the router.

Therefore the problem is a sequential decision problem, but not a formal online learning theorem statement.

2.2 Variables and notation

The core notation in the model section is mostly coherent:

- $z_t \in \{1, \dots, M\}$: latent regime;
- $g_t \in \mathbb{R}^{d_g}$: shared latent factor;
- $u_{t,k} \in \mathbb{R}^{d_\alpha}$: expert-specific latent state;
- $B_k \in \mathbb{R}^{d_\alpha \times d_g}$: loading matrix for expert k;
- $\Phi : \mathcal{X} \to \mathbb{R}^{d_\alpha \times d_y}$: feature map;
- $\alpha_{t,k} = B_k g_t + u_{t,k}$: latent reliability vector;
- $R_{m,k} \in \mathbb{S}^{d_y}_{++}$: emission covariance.

Well-definedness. The expression

$$\Phi(x_t)^\top \alpha_{t,k}$$

is dimensionally valid because $\Phi(x_t)^\top \in \mathbb{R}^{d_y \times d_\alpha}$ and $\alpha_{t,k} \in \mathbb{R}^{d_\alpha}$, so the output lies in $\mathbb{R}^{d_y}$; see `Paper/icml2026/Section/Approach.tex:211-230`.

Notation issue. The paper alternates between d_α and the implementation's d_ϕ for the idiosyncratic-state dimension. This is not mathematically wrong, but it creates friction when mapping paper to code. The implementation uses `d_phi` as the dimension of $u_{t,k}$ and of the feature vector returned by `feature_phi`; see `models/factorized_slds.py:68-110` and `models/router_model.py:5-9`.

2.3 Objective

The paper's operational decision rule is myopic. It computes predicted one-step costs, then combines them with an exploration score. The exploitation part is

$$\bar{C}^{\text{pred}}_{t,k} = \mathbb{E}[\psi(e^{\text{pred}}_{t,k}) \mid \mathcal{F}_t] + \beta_k$$

and the predictive gap is

$$\Delta_t(k) = \bar{C}^{\text{pred}}_{t,k} - \bar{C}^{\text{pred}}_{t,k^{\text{pred}}_t}, \qquad k^{\text{pred}}_t \in \arg\min_k \bar{C}^{\text{pred}}_{t,k}.$$

See `Paper/icml2026/Section/Approach.tex:306-318` and `Paper/icml2026/Section/AppendixParts/info_ga`

Important limitation. This is not a dynamic programming objective. It is a one-step predictive criterion plus an information heuristic. The paper is explicit about this point and does not claim Bayes optimal control; see `Paper/icml2026/Section/Approach.tex:299-304`.

2.4 Model

2.4.1 Regime dynamics

The regime dynamics are context-dependent:

$$\mathbb{P}(z_t = m \mid z_{t-1} = \ell, x_t) = \Pi_\theta(x_t)_{\ell m}.$$

The paper parameterizes $\Pi_\theta(x_t)$ by low-rank attention logits:

$$S(x_t) = \frac{1}{\sqrt{d_{\text{attn}}}} Q_\theta(x_t) K_\theta(x_t)^\top, \qquad \Pi_\theta(x_t)_{\ell m} = \frac{\exp(S_{\ell m}(x_t))}{\sum_j \exp(S_{\ell j}(x_t))}.$$

See `Paper/icml2026/Section/Approach.tex:135-163`.

Interpretation. **Explicitly stated.**The paper intends a learned context-dependent transition kernel.

Implicitly used. This assumes enough data and a stable enough training problem to identify the dependence of regime switching on context.

Missing. The paper never states what happens when transition parameters are untrained or weakly trained, even though the code permits exactly that.

2.4.2 Continuous latent dynamics

Conditioned on regime $z_t = m$:

$$g_t = A_m^{(g)} g_{t-1} + w_t^{(g)}, \qquad w_t^{(g)} \sim \mathcal{N}(0, Q_m^{(g)}),$$

$$u_{t,k} = A_m^{(u)} u_{t-1,k} + w_{t,k}^{(u)}, \qquad w_{t,k}^{(u)} \sim \mathcal{N}(0, Q_m^{(u)}).$$

See `Paper/icml2026/Section/Approach.tex:165-203`.

Comment. The paper shares $(A_m^{(u)}, Q_m^{(u)})$ across experts and uses expert-specific heterogeneity only through $u_{t,k}$ realizations and B_k; this is a modeling choice that improves statistical strength but imposes a strong inductive bias.

2.4.3 Emission model

The latent reliability of expert k is

$$\alpha_{t,k} = B_k g_t + u_{t,k},$$

and the residual emission is

$$e_{t,k} \mid (z_t = m, g_t, u_{t,k}, x_t) \sim \mathcal{N}\Big(\Phi(x_t)^\top \alpha_{t,k}, R_{m,k}\Big).$$

See `Paper/icml2026/Section/Approach.tex:205-239`.

Interpretation. **Explicitly stated.**The model is defined for residuals, not losses.

Inferred. This is because residuals can be negative and therefore admit a linear Gaussian state-space treatment, whereas losses generally do not.

Missing. The paper does not discuss misspecification when the true residual law is heavy-tailed, skewed, heteroskedastic in a way not captured by $R_{m,k}$, or nonlinear in $\Phi(x_t)$.

2.5 Inference / estimation

2.5.1 Filtering

The filter is described in algorithmic form in `Paper/icml2026/Section/AppendixParts/alg_router.tex:10-58`. The intended per-round recursion is:

1. update the registry $\mathcal{K}_t$ by adding current experts and pruning stale unavailable experts;
2. initialize entering experts with default idiosyncratic priors;
3. perform an IMM predictive step using $\Pi_\theta(x_t)$;
4. time-update the Gaussian beliefs of g_t and each stored $u_{t,k}$;
5. compute per-expert predictive residual laws;
6. choose an action via predicted cost and information ratio;
7. after observing the queried residual, update the joint state (g_t, u_{t,I_t}) by a Kalman correction;
8. project the updated covariance back to a factorized family by dropping posterior cross-covariances.

Mathematical status. The exact queried update is standard Kalman conditioning on the joint state. The approximation enters only at the projection step.

2.5.2 Cross-covariance and factorization

The appendix explicitly notes that even if the predictive covariance is block-diagonal, the posterior covariance after conditioning on the queried residual generally has a non-zero g–u_{I_t} cross-covariance; see `Paper/icml2026/Section/AppendixParts/derivations.tex:4-66`. This is important:

- the paper is aware that the exact posterior is not factorized;
- the factorized filter is therefore a projection approximation, not an exact Bayes update.

Critique. This is the central approximation of the method, but the paper gives no quantitative control on the error induced by repeatedly discarding these cross-covariances. That is acceptable for an empirical methods paper, but not for a learning-theoretic claim.

2.5.3 Parameter learning

The paper's learning routine is optional MCEM; see `Paper/icml2026/Section/AppendixParts/alg_learning.te`
The algorithmic idea is:

- sample regime paths $z_{1:T}$ given current latent trajectories,
- sample $g_{1:T}$ given $z_{1:T}$ and expert-specific trajectories,
- sample $u_{1:T,k}$ expertwise,
- update (A, Q, B, R) by weighted moment matching / regression,
- update the transition model parameters θ from expected regime transition counts.

Assessment. The algorithm is plausible and the code largely implements it. What is missing is a statement of convergence assumptions or identifiability.

2.6 Decision rule

The decision rule is an IDS-inspired information ratio:

$$I_t \in \arg\min_{k \in \mathcal{E}_t} \frac{\Delta_t(k)^2}{\mathrm{IG}_t(k)},$$

where

$$\mathrm{IG}_t(k) = \mathcal{I}\Big((z_t, g_t); e_{t,k}^{\mathrm{pred}} \mid \mathcal{F}_t\Big).$$

The decomposition

$$\mathcal{I}((z_t, g_t); e) = \mathcal{I}(z_t; e) + \mathcal{I}(g_t; e \mid z_t)$$

is given in `Paper/icml2026/Section/AppendixParts/info_gain.tex:7-34` and `Paper/icml2026/Section/Appe`

Shared-factor term. The closed-form term

$$\frac{1}{2} \log\det\Big(I + H_{t,k} \Sigma_{g,t|t-1}^{(m)} H_{t,k}^{\top} (S_{t,k}^{(m)})^{-1}\Big)$$

is correct for a linear Gaussian channel.

Mode-identification term. The paper estimates $\mathcal{I}(z_t; e_{t,k}^{\mathrm{pred}} \mid \mathcal{F}_t)$ by Monte Carlo under a Gaussian mixture model. This is technically sound as an estimator of the quantity it defines.

Theoretical status. **Explicitly stated.**The paper states this is a pragmatic exploration heuristic, not an optimality guarantee; see `Paper/icml2026/Section/Approach.tex:299-304`.

Critical point. This honesty matters. The paper should not be read as proving an IDS guarantee. It does not. It only borrows the information-ratio shape.

2.7 Computational complexity

The paper mentions scalability motivations, but it does not provide a clean complexity statement. The practical complexity depends on:

- the size of the active registry,
- the latent dimensions d_g and d_α,
- the number of regimes M,
- the Monte Carlo budget for $I(z_t; e)$.

What is clear. The factorized representation avoids storing a full joint covariance across all experts, which would scale poorly with registry size.

What is missing. There is no explicit asymptotic runtime or memory claim in the paper, so the reader cannot evaluate the scalability gain precisely from the manuscript alone.

3 Assumptions and Hidden Assumptions

3.1 Explicit assumptions

The following assumptions are stated or directly encoded in the formal model:

- **Exogeneity of the data stream:** past routing actions do not affect the law of $(x_t, \mathcal{E}_t, y_t)$ beyond the censoring of observations; see `Paper/icml2026/Section/Approach.tex:76-89`.
- **Conditional Gaussianity:** both latent dynamics and emissions are linear Gaussian given the regime; see `Paper/icml2026/Section/Approach.tex:174-239`.
- **Positive-definite noise covariances:** $Q_m^{(g)}$, $Q_m^{(u)}$, and $R_{m,k}$ are positive definite.
- **Factorized filtering approximation:** posterior cross-covariances are discarded after each queried update; see `Paper/icml2026/Section/Approach.tex:250-257` and `Paper/icml2026/Section/Appenc`
- **Shared idiosyncratic dynamics parameters across experts:** $(A_m^{(u)}, Q_m^{(u)})$ do not depend on k.

3.2 Implicit assumptions

The method also relies on several assumptions that are not made explicit enough:

- **Persistent expert identities.** The meaning of re-entry depends on an expert ID corresponding to the same forecasting mechanism over time. This is used operationally but not formally discussed.
- **Identifiability of the decomposition** $B_k g_t + u_{t,k}$. Without additional constraints, there can be tradeoffs between the shared and idiosyncratic components.
- **Stable enough dynamics.** The filter and EM updates become difficult to interpret if $A_m^{(g)}$ or $A_m^{(u)}$ imply explosive latent trajectories.
- **Availability process does not induce confounding.** If availability is itself coupled to latent difficulty in a way that violates exogeneity, the probabilistic interpretation changes.
- **Feature map adequacy.** The model assumes $\Phi(x_t)$ captures the context dependence of residuals well enough for linear-Gaussian modeling to be informative.

3.3 Missing assumptions

Several assumptions are needed for stronger claims but are absent:

- conditions for identifiability of $(B_k, g_t, u_{t,k})$;
- consistency or convergence conditions for MCEM under censoring;
- approximation-error control for repeated posterior projection;
- assumptions under which the IDS-inspired rule improves long-run performance;
- assumptions under which pruning and re-entry preserve not just retained marginals, but decision quality for re-entering experts.

4 Theory Audit

4.1 Evaluation of the formal claims

4.1.1 Problem formulation

The problem formulation is precise enough to define the stochastic process and the one-step Bayes benchmark. However, it does not support a specialist-level online learning claim in the regret sense, because no comparator or regret is defined.

Diagnosis. The paper is better described as a model-based sequential decision method under partial feedback than as a learning-theoretic online-learning paper.

4.1.2 Proposition on cross-update

Proposition `prop:cross_update` states that, conditional on fixed regime and joint Gaussianity, observing the queried residual changes the predictive mean of another expert if and only if the predictive cross-covariance is non-zero; see `Paper/icml2026/Section/Approach.tex:259-278` and `Paper/icml2026/Section/AppendixParts/proofs.tex:10-38`.

Assessment. This proposition is correct. It is a direct consequence of Gaussian conditioning:

$$\mathbb{E}[X \mid Y = y] = \mu_X + \Sigma_{XY}\Sigma_Y^{-1}(y - \mu_Y).$$

Limitation. It is an elementary structural fact, not a performance theorem.

4.1.3 Pruning proposition

Proposition `prop:invariance` states that pruning experts from the registry does not change the retained marginals because pruning is defined by marginalization; see `Paper/icml2026/Section/Approach.tex:384-` and `Paper/icml2026/Section/AppendixParts/proofs.tex:44-86`.

Assessment. The proposition is correct but tautological. It proves exactly what its definition encodes.

What it does not prove. It does not show that re-entered experts are well initialized, only that retained experts are not perturbed by pruning.

4.1.4 Birth coupling proposition

Proposition `prop:transfer` states that under the factorized predictive belief,

$$\mathrm{Cov}(\alpha_{t,j}, \alpha_{t,k} \mid \mathcal{F}_t, z_t = m) = B_j \Sigma_{g,t|t-1}^{(m)} B_k^\top,$$

so shared-factor uncertainty induces coupling across experts; see `Paper/icml2026/Section/Approach.tex:449-46` and `Paper/icml2026/Section/AppendixParts/proofs.tex:92-117`.

Assessment. This proposition is correct as stated for the latent reliability vectors $\alpha_{t,j}, \alpha_{t,k}$.

Important caveat. It is *not* the proposition that justifies queried update transfer at the residual level. That role belongs to `prop:cross_update`. The paper later mis-cites this point in the synthetic results.

4.1.5 IDS-inspired exploration

The paper's mutual-information construction is technically coherent and the shared-factor term is derived correctly. The mode-identification term is estimated by Monte Carlo because the predictive residual law is a Gaussian mixture; see `Paper/icml2026/Section/AppendixParts/info_gain.tex:31-34` and `Paper/icml2026/Section/AppendixParts/info_gain.tex:204-260`.

Assessment. As a heuristic exploration score under the model, this is fine.

What is not justified. There is no proof that minimizing the resulting ratio improves cumulative cost, minimizes Bayes regret, or satisfies any IDS theorem. The paper is careful in its wording, but the framing is still stronger than the theorem layer.

4.2 Proof gaps or questionable steps

1. **No approximation analysis for factorization.** The main computational approximation is the repeated projection to a factorized posterior. The paper acknowledges this, but gives no bound on how much predictive distributions or information scores may drift from the exact posterior.
2. **Latent-to-residual transfer is not fully unpacked.** The paper correctly notes that latent shared-factor covariance induces residual cross-covariance through $\Phi(x_t)^\top B_k$, but it does not spell out the minimal non-degeneracy conditions under which this actually affects the queried conditional mean for another expert.
3. **No identifiability discussion.** The same residual variation can often be explained by g_t or by $u_{t,k}$. This matters both conceptually and for MCEM, but the paper does not discuss it.

4.3 Claims correct only under stronger assumptions

- **Context-aware switching.** This only means something if transition parameters are trained or otherwise intentionally specified. In the actual synthetic and Melbourne configs, EM is disabled while the transition mode is attention-based, so the model uses random transition parameters unless trained elsewhere.
- **General vector output $d_y \geq 1$.** The notation supports it, but all provided environments and experiments are scalar-output. The claim is formal generality, not demonstrated generality.
- **Cross-expert transfer in practice.** The theoretical mechanism requires non-trivial shared loadings and a feature map that does not suppress the shared channel. Without these, transfer can vanish.

4.4 Possible counterexamples or failure modes

- If $B_k = 0$ for all k, then the shared factor plays no role and the model collapses to independent expert-specific processes.
- If $\Phi(x_t)$ is zero or orthogonal to the shared component, then even non-zero latent shared covariance need not produce residual-level transfer.

- If availability depends adversarially on latent difficulty, the exogeneity assumption fails and the interpretation of partial feedback as pure censoring becomes invalid.
- If the true residual law is strongly non-Gaussian, the information scores may be badly calibrated.

5 Implementation Walkthrough

5.1 Repository structure

The implementation relevant to the paper is distributed across:

- `slds_imm_router.py`: experiment driver and router construction;
- `models/factorized_slds.py`: proposed model implementation;
- `router_eval.py`: interaction loops with environments;
- `plot_utils.py`: aggregation, masking, and reporting;
- `environment/synthetic_env.py`: synthetic benchmark;
- `environment/etth1_env.py`: time-series data environment also used for FRED-like and Melbourne-like experiments;
- `models/linucb_baseline.py` and `models/neuralucb_baseline.py`: baseline implementations;
- `config/exp_synthetic_1.yaml`, `config/config_melbourne.yaml`, and `config/exp_FRED.yaml`: experiment protocols.

5.2 Entry points

The proposed method is instantiated in `slds_imm_router.py:1247-1575`. The driver builds one or more `FactorizedSLDS` instances depending on:

- feedback mode,
- transition mode,
- exploration mode,
- inclusion of the no-g_t ablation.

Offline EM, when enabled, is dispatched from `slds_imm_router.py:1980-2104` and parameter copies are shared across paired router instances in `slds_imm_router.py:2295-2367`.

5.3 Data flow

At evaluation time, `router_eval.py` does the following:

1. fetch context and availability at time t,
2. call the router's `select\_expert`,
3. obtain either residual or loss observations,
4. record cost,
5. update the router.

See `router_eval.py:486-554`.

For `FactorizedSLDS`, the observation passed to the model is the residual, not the loss, in `router_eval.py:216-243`. This matches the paper.

5.4 Model components

5.4.1 Observation model in code

The class docstring in `models/factorized_slds.py:48-66` implements the same residual emission model as the paper:

$$e_{t,k} = \phi(x_t)^\top (B_k g_t + u_{t,k}) + \varepsilon.$$

The implementation only supports residual observations: `models/factorized_slds.py:107-108` and `models/factorized_slds.py:259-264`.

5.4.2 Feature map

The default feature map is identity: `models/router_model.py:5-9`.

Consequence. For the provided experiments, $\Phi(x_t)$ is effectively a vectorized identity map on the context features. This is simpler than the general matrix-valued Φ notation in the paper.

5.4.3 Transition model

The implementation supports:

- an attention-based transition parameterization,
- a linear transition parameterization,
- an optional neural transition model.

The default is `transition_mode="attention"` and `transition_init="random"` in `models/factorized_slds.py`
The decision-time transition matrix is produced by `\_context\_transition` in `models/factorized_slds.py:741`

Critical implementation fact. If EM is disabled and no trained transition model is loaded, then `\_context\_transition` uses random initialized parameters. This is not a minor implementation detail; it affects the meaning of the paper's "context-aware switching" claim in the no-EM experiments.

5.5 Training loop

The EM implementation in `models/factorized_slds.py:3390-4050` broadly matches the appendix:

- latent regime sampling,
- Kalman sampling for g_t,
- Kalman sampling for each $u_{t,k}$,
- M-step updates for A_g, Q_g, A_u, Q_u, B_k, $R_{m,k}$,
- gradient-based transition training.

Positive note. This is one of the stronger paper-to-code alignments in the repository.

5.6 Inference / prediction

Predictive statistics are computed in `models/factorized_slds.py:1377-1459`. The code:

- computes per-mode predictive means and covariances,
- computes expected loss under each mode,

- computes the shared-factor mutual-information term,
- returns per-expert predicted cost and information gain.

Mode-identification information is estimated in `models/factorized_slds.py:1535-1613`. IDS selection is implemented in `models/factorized_slds.py:1679-1786`.

5.7 Evaluation procedure

Evaluation masking occurs in `plot_utils.py:1085-1155`: costs and choices before `em\_tk` are replaced by `NaN`. Reported average costs are then computed with `np.nanmean` in `plot_utils.py:1291-1335`. Constant expert baselines are also recomputed starting from `t\_start = em\_tk + 1` in `plot_utils.py:1267-129`

Interpretation. This means the reported summaries for experiments with EM warm-up are *post-warmup* averages, not full-horizon averages, unless explicitly stated otherwise.

6 Paper-to-Code Alignment

Paper component	Location in paper	Code file/function	Status	Explanation
Sequential partial-feedback routing	`Approach.tex:45-73`	`router_eval.py:216-243`, `router_eval.py:486-554`	Match	The code passes only the queried residual to `FactorizedSLDS` in partial mode, matching the paper's observation model.
Residual state-space model	`Approach.tex:205-239`	`models/factorized_slds.py:48-66`	Match	The implemented observation model is residual-based and uses the same shared plus idiosyncratic decomposition.
Context-dependent regime transitions	`Approach.tex:135-163`	`models/factorized_slds.py:741-763`	Partial match	The code implements context-dependent transitions, but in some published experiments the parameters are random because EM is disabled.
Predictive cost computation	`info_gain.tex:128-161`	`models/factorized_slds.py:1377-1459`	Match	The code computes per-mode predictive moments and expected loss exactly in the spirit of the appendix.
Shared-factor IG term	`info_gain.tex:188-202`	`models/factorized_slds.py:1431-1443`	Match	The scalar and multivariate log-det forms are implemented.
Mode-identification term	`info_gain.tex:204-260`	`models/factorized_slds.py:1535-1613`	Match	Monte Carlo KL-to-mixture estimator is implemented with stable log-sum-exp.

Paper component	Location in paper	Code file/function	Status	Explanation
IDS-inspired routing rule	`Approach.tex:333-340`	`models/factorized_slds.py:1679-1738`	Match; extra heuristics	The code adds score clipping and greedy fallbacks not documented in the paper.
Queried Kalman update and factorized projection	`alg_router.tex:39-58`	`models/factorized_slds.py:1980-2068`	Match	The code updates the joint state (g_t, u_{t,I_t}), then stores only block marginals.
Registry pruning	`Approach.tex:362-405`	`models/factorized_slds.py:2385-2413`	Match	Stale unavailable experts are dropped after `Delta_max`; re-entry restarts from default priors.
Offline MCEM	`alg_learning.tex:9-12`	`models/factorized_slds.py:3390-4057`	Match	The code contains a genuine latent-variable EM-style procedure broadly consistent with the appendix.
Synthetic experiment design	`Experiments.tex:55-123`	`environment/synthetic_env.py:547-582`, `config/exp_synthetic_1.yaml:44-79`	Mismatch	The code uses a latent model-matched residual generator with `shared_dim: 2`, not the additive expert-noise simulator with $d_g = 1$ described in the paper.
Synthetic warmup of 100 steps	`Experiments.tex:121-123`	`router_eval.py:514-554`, `config/exp_synthetic_1.yaml:98-100`	Mismatch	The published synthetic config disables EM and the interaction starts immediately from $t = 1$.
Melbourne no-EM claim	`Experiments.tex:158-162`	`config/config_melbourne.yaml:123-153`	Match; config level	EM is indeed disabled in the Melbourne config. However, the transition model remains attention-based and therefore untrained unless separately learned.
FRED evaluation protocol	Appendix experiment text	`config/exp_FRED.yaml:156-168`, `plot_utils.py:1085-1155`	Match	The code uses offline full-feedback EM on the prefix and reports post-prefix averages after masking the warmup.
NeuralUCB baseline	`AppendixParts/Experiments.tex:83-110`	`models/neural_ucb_baseline.py:5-25`	Mismatch	The paper describes gradient-feature NeuralUCB; the code explicitly implements a NeuralLinear-style approximation.

Paper component	Location in paper	Code file/function	Status	Explanation
General $d_y \geq 1$ support	`Approach.tex:13-15`, `Approach.tex:211-230`	`models/router_model.py:5-9`, environments	Under / weakly supported	The code can accept vector residuals in principle, but all provided experiments are scalar-output and the default feature map is vector-valued identity.

7 Issues Found

7.1 Critical errors

7.1.1 Synthetic experiment in code does not match the paper's stated simulator

Status: Critical mismatch

Paper statement. The paper says the synthetic experts are

$$\hat{y}_{t,k} = 0.8y_{t-1} + b_k + \varepsilon_{t,k},$$

with regime-dependent correlated additive noises constructed from two shared scalar noises and idiosyncratic noise; see `Paper/icml2026/Section/Experiments.tex:67-90`. It also states $d_g = 1$ in `Paper/icml2026/Section/Experiments.tex:119-123`.

Code reality. The synthetic config instead sets `shared_noise_mode: "model_match"` and `shared_dim: 2` in `config/exp_synthetic_1.yaml:44-46` and `config/exp_synthetic_1.yaml:78-79`. The environment then generates latent shared factors g_t, latent expert-specific states u_t, and residuals via

```
residuals[t] = x_t * (loadings @ g_t + u_t) + noise,
```

implemented in `environment/synthetic_env.py:571-577`. The actual expert prediction is

$$\hat{y}_{t,k} = y_t + \texttt{residual}_{t,k}$$

in `environment/synthetic_env.py:841-846`.

Why this matters. This is not a cosmetic mismatch. The code's simulator is substantially closer to the proposed model class than the paper's additive-intercept simulator. That can materially bias the benchmark in favor of L2D-SLDS.

Minimal fix. Either:

- rewrite the paper to describe the actual latent residual generator, or
- change the experiment code to the stated additive-noise simulator and rerun the results.

7.1.2 Context-dependent transitions are random and untrained in synthetic and Melbourne runs

Status: Critical paper-code mismatch

Paper statement. The method is presented as using context-dependent regime transitions $\Pi_\theta(x_t)$ with learned low-rank attention logits; see `Paper/icml2026/Section/Approach.tex:135-163`.

Code fact. The `FactorizedSLDS` defaults are `transition_mode="attention"` and `transition_init="random"` in `models/factorized_slds.py:114-115`. If no trained transition model is present, `\_context\_transition` constructs transition scores from those parameters in `models/factorized_slds.py:741-763`. The synthetic config uses `transition_mode: "attention"` and `em_enabled: false` in `config/exp_synthetic_1.y` The Melbourne config does the same in `config/config_melbourne.yaml:123-153`.

Consequence. In those experiments, the purported context-aware transition kernel is not learned. It is simply a random function of context unless some external initialization path exists. I found no such training path for those runs.

Why this matters. This invalidates the natural reading that the reported no-EM results are using a meaningful learned context-driven switching mechanism.

Minimal fix. If EM is disabled, default to a fixed homogeneous transition matrix or disclose that the transition scores are random and not learned.

7.1.3 FRED uses full-feedback offline EM and masked post-prefix reporting

Status: Critical evaluation-protocol mismatch

Code fact. The FRED config sets `em_enabled: true`, `em_offline_full_feedback: true`, and `em_tk: 2000` in `config/exp_FRED.yaml:166-168`. The driver forces `router.feedback_mode = "full"` during offline EM in `slds_imm_router.py:1983-2069`. The evaluation code masks all costs before `em_tk` to `NaN` in `plot_utils.py:1085-1155`, and then reports `np.nanmean` in `plot_utils.py:1291-1335`.

Consequence. The reported averages in such experiments are not full-horizon online partial-feedback performance. They are post-warmup averages after a full-feedback parameter estimation phase on the initial prefix.

Why this matters. This is a different experimental protocol. It may be perfectly legitimate, but it must be stated plainly because it changes what the results mean.

7.1.4 The paper's NeuralUCB baseline is not the implementation

Status: Critical baseline mismatch

Paper statement. The appendix describes NeuralUCB using gradient features $h_{t,k} = \nabla_\omega f_\omega(x_t, k)$ and a single Gram matrix in parameter space; see `Paper/icml2026/Section/AppendixParts/Experiments.tex:8`

Code fact. The implementation explicitly states: "This is a 'NeuralLinear' approximation to NeuralUCB" in `models/neuralucb_baseline.py:22-25`. It uses:

- a one-hidden-layer ReLU MLP embedding,
- per-expert ridge matrices in embedding space,
- SGD updates to the embedding.

See `models/neuralucb_baseline.py:70-83` and `models/neuralucb_baseline.py:165-192`.

Consequence. The baseline is mislabeled. This matters because NeuralUCB's theoretical and empirical behavior is not the same as a NeuralLinear approximation.

7.2 Moderate concerns

7.2.1 Synthetic warmup described in paper is not implemented in the published config

Status: Moderate mismatch

Paper statement. The synthetic experiment section says: "We simply run a small warmup of 100 steps before running L2D-SLDS and UCBs" in `Paper/icml2026/Section/Experiments.tex:121-123`.

Code fact. The synthetic config disables EM in `config/exp_synthetic_1.yaml:98`. The main interaction loop starts from $t = 1$ in `router_eval.py:514-554`. I did not find a separate 100-step synthetic warmup path for the published config.

Interpretation. Either the text is outdated or the config is.

7.2.2 Synthetic multi-seed reporting uses a fixed data seed by default

Status: Moderate reporting issue

Code fact. The synthetic environment hard-codes `DATA_GENERATION_SEED = 42` unless `data_seed` is explicitly provided; see `environment/synthetic_env.py:3-8` and `environment/synthetic_env.py:44-50`. The helper script `scripts/run_exp_synth_seeds.sh:34-39` only changes `environment.seed`, not `data_seed`.

Consequence. By default, multi-seed synthetic runs vary learner randomness while keeping the underlying synthetic time series and expert behaviors fixed.

Why this matters. That is acceptable if disclosed. It is misleading if the paper suggests the error bars are over independent environment draws.

7.2.3 Undocumented initialization heuristics affect Melbourne

Status: Moderate method-reporting gap

Code fact. The Melbourne config uses:

- `init_R_mode: empirical_expert`,
- `init_state_from_residuals: u`,

in `config/config_melbourne.yaml:112-121`. These heuristics are implemented in `slds_imm_router.py:1827-19`

Consequence. The model is not being evaluated only with the generic priors described in the paper. It receives data-dependent initialization help.

Why this matters. Again, this may be sensible engineering, but it belongs in the paper if the reported results depend on it.

7.2.4 Diagnostic sample count can change the actual routing behavior

Status: Moderate implementation quirk

Code fact. When diagnostics are enabled, `select_expert` precomputes `iz_scores` using `exploration_diag_sa` and then reuses those scores in the action selection path; see `models/factorized_slds.py:1820-1836`.

Consequence. The number of diagnostic Monte Carlo samples can affect the selected actions, not just the logs. This is undocumented.

7.3 Minor issues / style / clarity issues

1. **Wrong proposition cited in synthetic analysis.** The synthetic results section cites `prop:transfer` for query-time information transfer in `Paper/icml2026/Section/Experiments.tex:143-144`. The relevant result is `prop:cross_update`, not `prop:transfer`.
2. **Melbourne text/table inconsistency.** Table values show NeuralUCB as 5.88 ± 0.04 and Random as 6.71 ± 0.03 in `Paper/icml2026/Section/Experiments.tex:177-182`, but the narrative in `Paper/icml2026/Section/Experiments.tex:194-196` says NeuralUCB is 6.71 ± 0.03.
3. **Incorrect percentage arithmetic.** The increase from 5.69 to 5.78 is about 1.58%, not 16%.
4. **Indexing inconsistency in synthetic regimes.** The text defines $z_t \in \{1, 2\}$ in `Paper/icml2026/Section/Ex` but the figure caption refers to "Regime-0" in `Paper/icml2026/Section/Experiments.tex:37-42`.
5. **README command is stale.** The README points to `config/config_synth_paper.yaml` in `README.md:24`, but that file does not exist in the repository root.

8 Concrete Fixes

8.1 Paper fixes

- Rewrite the synthetic experiment description to match the actual code, or replace the code with the stated simulator and regenerate the table and figure.
- Correct the Melbourne table narrative, the 16% arithmetic mistake, the regime indexing inconsistency, and the wrong proposition citation.
- State explicitly when experiments use offline EM, whether it uses full feedback, and whether reported averages exclude the warmup prefix.
- Replace "NeuralUCB" by "NeuralLinear-style baseline" unless the true algorithm is implemented.

- Add a short implementation-details paragraph documenting empirical initialization of R and initial state covariances.

8.2 Code fixes

- If `em_enabled: false`, set a deterministic fallback transition matrix instead of random attention weights, or require explicit transition parameters in the config.
- Add a paper-faithful synthetic mode implementing the exact equations in the manuscript.
- Decouple diagnostics from decision-making by ensuring `exploration_diag_samples` never overrides the action-path Monte Carlo budget.
- Update the README command to a valid config path.

8.3 Experimental fixes

- Rerun synthetic experiments under the exact simulator stated in the paper.
- Report both full-horizon and post-warmup averages whenever EM warmup is used.
- For synthetic seed sweeps, vary both the learner seed and the data seed unless the goal is specifically to condition on one environment draw.
- Either implement the baseline described as NeuralUCB or relabel the existing results.

8.4 Theoretical fixes

- Narrow the framing: present the method as a model-based routing heuristic under partial feedback, not as a learning-theoretic result unless a regret statement is added.
- Add a discussion of identifiability and approximation error from posterior projection.
- State explicitly the minimal conditions under which shared-factor covariance yields residual-level transfer.

9 Final Verdict

9.1 Is the problem formulation clear?

Mostly yes. The paper defines the sequential routing problem, the observables, the latent variables, the cost, and the one-step benchmark clearly. The main gap is that it does not define an online-learning benchmark such as regret.

9.2 Is the method mathematically sound?

The core state-space model and filter are mathematically coherent. The main approximation, namely the projection to a factorized posterior, is transparent and honestly described. The weakest mathematical points are the absence of identifiability discussion and the lack of any bound on approximation error or decision-quality loss from the projection.

9.3 Are the theoretical claims justified?

Only partially. The structural propositions are correct but limited. The paper does not justify stronger claims about online optimality, regret, or asymptotic performance.

9.4 Does the code faithfully implement the paper?

The core `FactorizedSLDS` implementation is substantially faithful to the paper's filtering and MCEM description. The *experimental protocols* are not faithfully aligned. The major paper-code mismatches are:

- the synthetic simulator,
- the untrained random transition model in no-EM runs,
- the FRED full-feedback offline EM plus masked evaluation,
- the mislabeled NeuralUCB baseline.

9.5 Top three risks that could invalidate the results

1. **Synthetic benchmark mismatch.** The code appears more model-matched to L2D-SLDS than the paper discloses.
2. **Random transitions in no-EM experiments.** The claimed context-aware switching mechanism is not meaningfully learned in synthetic and Melbourne under the published configs.
3. **Protocol and baseline mismatches.** Full-feedback EM on FRED and the mislabeled NeuralUCB baseline weaken the credibility of comparative empirical claims.

Bottom line. As a modeling paper, the work has a coherent core. As a rigorous paper-code package, it currently has significant specification and evaluation problems that must be corrected before the empirical claims can be treated as fully reliable.